EXPERIMENT - 7

AIM: Build an Artificial Neural Network (ANN) using Keras/TensorFlow using churn modelling dataset.

OUTPUT:

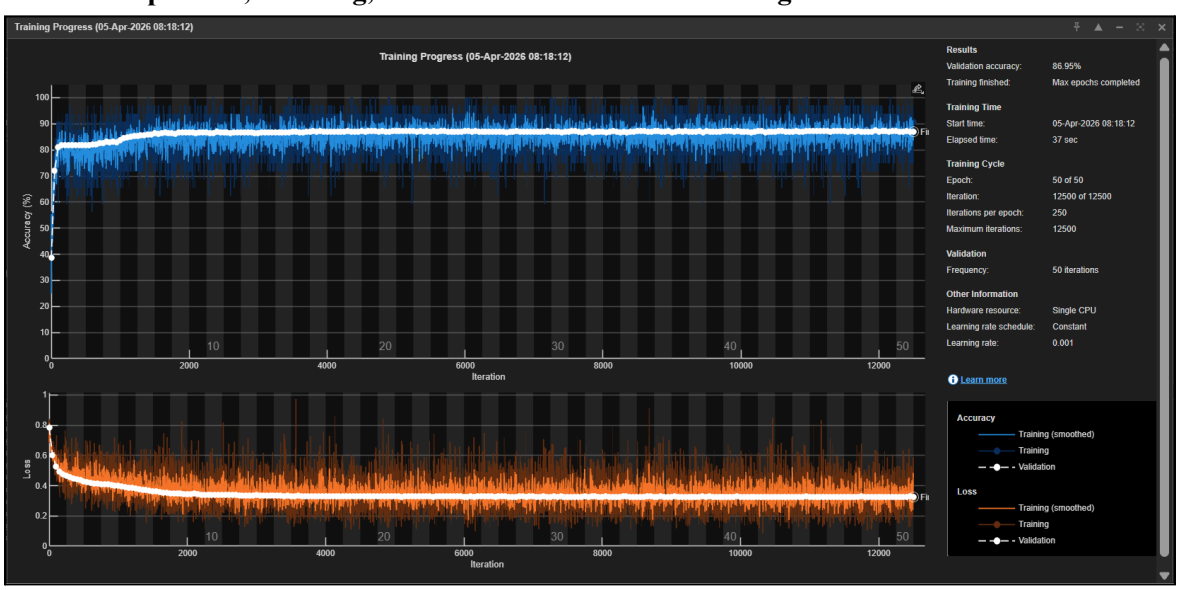
Load and Explore Dataset

```
>> experiment_7
    CreditScore    Geography      Gender      Age    Tenure     Balance      NumOfProducts    HasCrCard    IsActiveMember    EstimatedSalary    Exited
    ___________    _________    __________    ___    ______    __________    _____________    _________    ______________    _______________    ______

        619        {'France'}    {'Female'}    42       2            0           1               1             1              1.0135e+05         1
        608        {'Spain' }    {'Female'}    41       1        83808           1               0             1              1.1254e+05         0
        502        {'France'}    {'Female'}    42       8    1.5966e+05          3               1             0              1.1393e+05         1
        699        {'France'}    {'Female'}    39       1            0           2               0             0                  93827          0
        850        {'Spain' }    {'Female'}    43       2    1.2551e+05          1               1             1                  79084          0
```

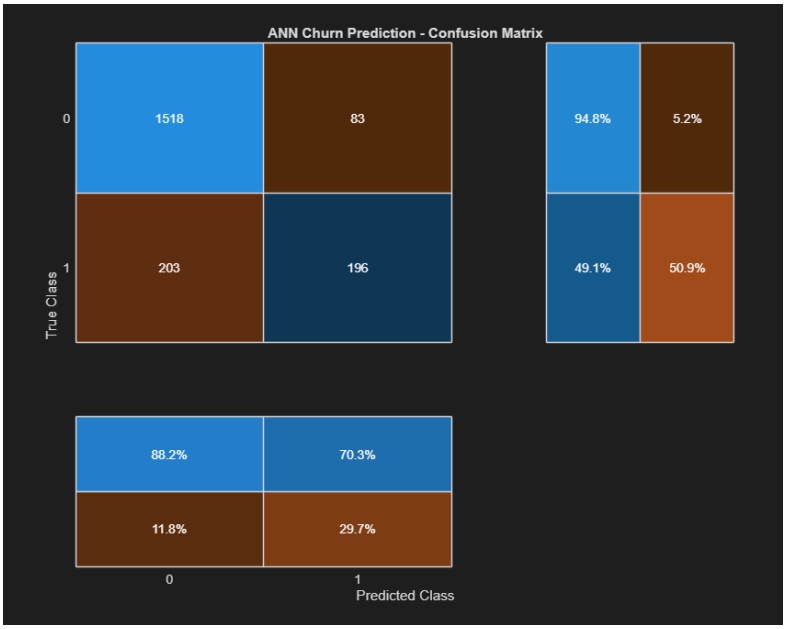
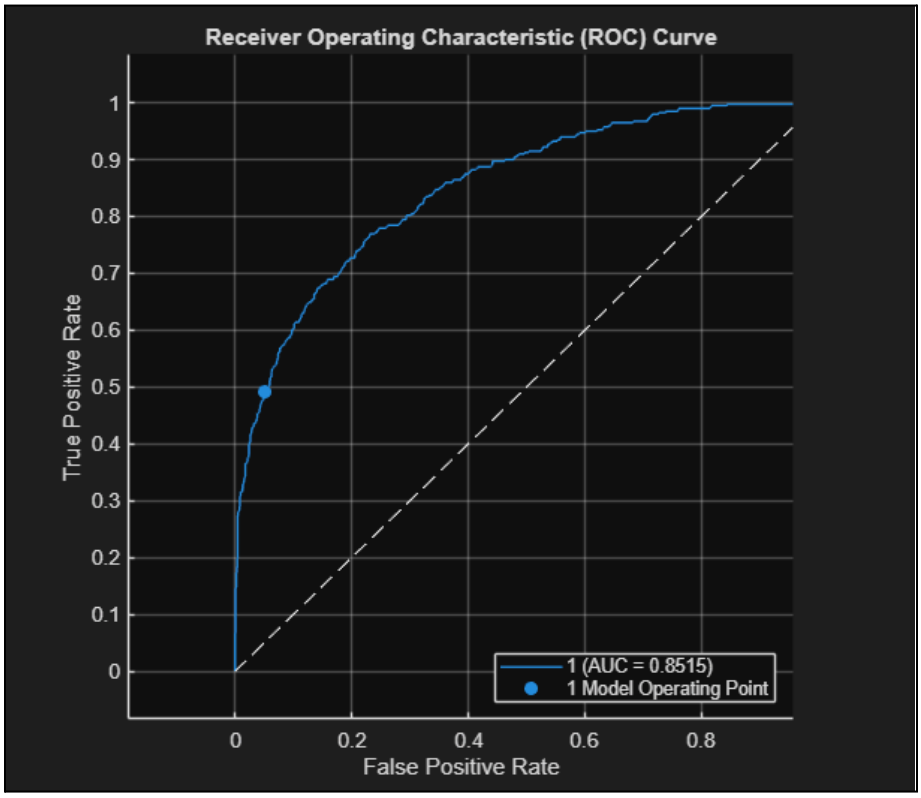
Define Neural Network Architecture

```
X_test =

   -1.0445   -0.9954   -0.5783    1.7245   -1.0941   -1.4347    1.3823   -1.2288    0.8046    0.6457    0.9701   -1.4933
    0.8456    1.0044   -0.5783   -0.5798    0.9139    0.1983    1.0373   -1.2288    0.8046    0.6457    0.9701    1.2330
   -1.4473   -0.9954   -0.5783    1.7245   -1.0941   -0.0899   -0.3424   -1.2288   -0.9161    0.6457   -1.0307    0.3275
   -0.7036   -0.9954    1.7291   -0.5798    0.9139    0.1983    0.3474   -0.0998    0.8046   -1.5486    0.9701    1.3582
   -1.9120    1.0044   -0.5783   -0.5798   -1.0941    1.1589    1.0373    0.7374   -0.9161   -1.5486   -1.0307    1.4144
    1.8991    1.0044   -0.5783   -0.5798   -1.0941    0.9668   -1.0323    0.8798   -0.9161   -1.5486   -1.0307    1.6421
    1.3000   -0.9954    1.7291   -0.5798   -1.0941   -0.6662   -0.3424    0.5272    0.8046    0.6457    0.9701    0.4600
   -0.6727   -0.9954    1.7291   -0.5798    0.9139   -0.2820    0.0025    1.1150    0.8046   -1.5486   -1.0307   -0.2385
   -1.4369   -0.9954   -0.5783    1.7245   -1.0941    2.5998   -0.3424   -1.2288   -0.9161    0.6457   -1.0307   -1.7156
    0.9075   -0.9954    1.7291   -0.5798    0.9139    1.8313   -1.0323    0.9175    4.2461    0.6457   -1.0307   -1.2499
    1.3103    1.0044   -0.5783   -0.5798   -1.0941   -0.6662   -1.0323   -1.2288   -0.9161    0.6457   -1.0307    0.6332
   -1.4886    1.0044   -0.5783   -0.5798   -1.0941   -0.4741   -0.3424    0.2205   -0.9161    0.6457    0.9701    1.0300
   -0.0323   -0.9954   -0.5783    1.7245   -1.0941    0.4865    0.0025   -1.2288    2.5253    0.6457    0.9701    1.2908
   -1.3027    1.0044   -0.5783   -0.5798   -1.0941   -0.2820    1.7272   -1.2288    0.8046    0.6457   -1.0307    0.1655
    1.2690   -0.9954   -0.5783    1.7245    0.9139    0.1983    1.3823    0.4214   -0.9161   -1.5486    0.9701   -0.6188
   -0.7036   -0.9954    1.7291   -0.5798    0.9139   -0.6662   -1.3773    0.1985   -0.9161    0.6457    0.9701   -1.5691
   -1.7675   -0.9954    1.7291   -0.5798    0.9139   -0.3780    1.3823    0.2610   -0.9161    0.6457   -1.0307   -0.0116
    0.3602   -0.9954    1.7291   -0.5798    0.9139   -0.8583   -0.6874    0.2241   -0.9161   -1.5486    0.9701   -0.6452
   -1.1581   -0.9954    1.7291   -0.5798    0.9139    0.0062    0.6924    0.5053    0.8046    0.6457   -1.0307   -1.2698
    0.3498   -0.9954    1.7291   -0.5798   -1.0941    0.8707    1.7272    0.7994   -0.9161    0.6457    0.9701    1.7076
   -0.2595   -0.9954    1.7291   -0.5798    0.9139    1.6392   -1.7222    1.1544   -0.9161    0.6457   -1.0307   -0.9285
    1.8371    1.0044   -0.5783   -0.5798   -1.0941    0.1983    0.6924   -1.2288    0.8046    0.6457   -1.0307    1.2416
    1.4343   -0.9954    1.7291   -0.5798    0.9139    0.0062    0.6924    0.7745    0.8046    0.6457    0.9701    0.4233
   -0.6314   -0.9954    1.7291   -0.5798   -1.0941    1.0629    0.0025    1.0965    0.8046    0.6457    0.9701   -1.1355
    0.3292    1.0044   -0.5783   -0.5798   -1.0941    0.3904    0.0025    0.7909   -0.9161    0.6457    0.9701    1.6238
    0.4221    1.0044   -0.5783   -0.5798   -1.0941   -0.7623    0.0025   -0.5722   -0.9161    0.6457   -1.0307    0.4547
   -1.7055   -0.9954   -0.5783    1.7245   -1.0941   -1.7228    0.0025    0.5872   -0.9161    0.6457    0.9701   -0.8010
   -0.0013    1.0044   -0.5783   -0.5798    0.9139   -0.1859    0.0025    0.4878   -0.9161   -1.5486   -1.0307   -1.3175
   -1.1891    1.0044   -0.5783   -0.5798   -1.0941    0.8707    1.3823   -1.2288   -0.9161    0.6457   -1.0307    0.8672
    0.6803   -0.9954   -0.5783    1.7245    0.9139    0.2944    1.0373   -1.2288    0.8046    0.6457   -1.0307    1.4057
   -1.1478    1.0044   -0.5783   -0.5798    0.9139   -0.1859   -1.0323    0.8191   -0.9161    0.6457   -1.0307   -0.0256
   -0.1976   -0.9954    1.7291   -0.5798   -1.0941   -0.5701   -0.3424    0.7491   -0.9161   -1.5486   -1.0307    0.2190
```

Model Compilation, Training, and Performance Benchmarking



Confusion Matrix



Model Validation: Confusion Matrix and Classification Metrics